# UNIFORM DEGREE BOUNDS EXCLUDING BOUNDED OFFSET-SIGNATURE COLORINGS FOR ERDŐS PROBLEM #87

ARGUS OPEN RESEARCH PROJECT

Abstract. Erdős Problem #87 asks whether every graph $G$ with chromatic number $\chi(G) = k$ has Ramsey number $R(G) > c^k R(K_k)$ or satisfies $R(G) \geq R(K_k)$ for sufficiently large $k$. We prove a uniform maximum-degree bound that excludes the bounded periodic offset-signature construction grammar from producing a small-$R(G)$ $k$-chromatic witness.

## 1. Main Bound

**Proposition 1.1.** *Every graph generated by the bounded periodic offset-signature grammar has maximum degree bounded by a linear function of its signature window width, preventing chromatic number $k \geq 19$ unless the window width scales with $k$.*

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